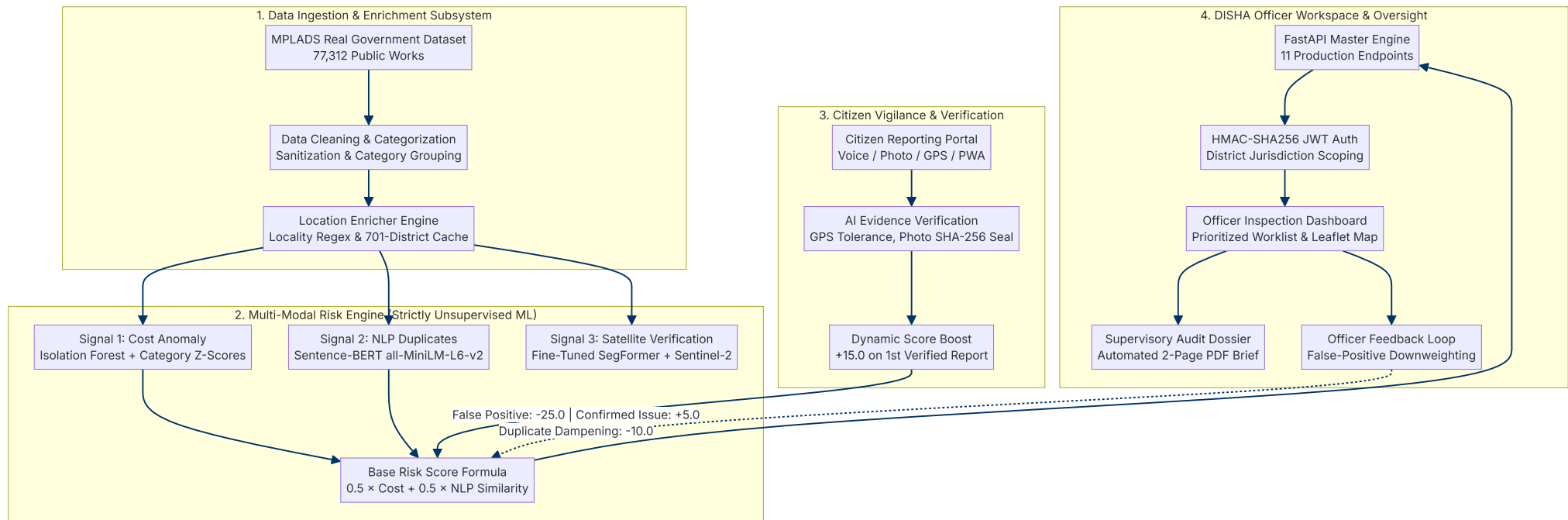


1. End-to-End Multi-Modal System Architecture

Four-subsystem pipeline connecting raw 77K records to officer audit actions and dynamic feedback loops

Master System Flow



DATA INGESTION

77,312 real MPLADS works. 3-tier geocode resolution (Locality, District centroid, Unavailable).

COST ANOMALY

Isolation Forest partitioned by work category. Generates 0–100 cost outlier risk score.

DUPLICATE DETECTION

384-dimensional Sentence-BERT embeddings. Cosine threshold $\geq 91\%$ prevents ghost funding.

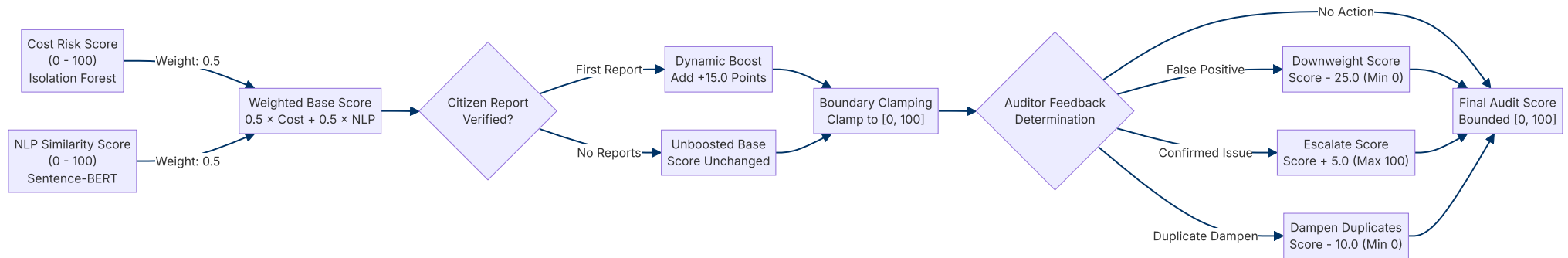
OFFICER FEEDBACK

Interactive false-positive dampening (-25 score) prevents systemic audit bias.

2. Risk Prioritization Scoring Formulation

Strict mathematical bounds [0, 100], auditable deterministic weightings, and statutory auditor overrides

Scoring Pipeline



STEP 1: BASE SCORE

$\text{Score_base} = 0.5 \times \text{CostScore} + 0.5 \times \text{NLPScore}$

STEP 2: CITIZEN BOOST

$\text{Score_boost} = \text{Score_base} + 15.0$ (1st report only)

STEP 3: NORMALIZATION

$\text{Score_norm} = \min(100.0, \max(0.0, \text{Score_boost}))$

STEP 4: STATUTORY OVERRIDE

FP: -25.0 | Confirmed: +5.0 | Dampen: -10.0

REVERSE-ENGINEERED FIT

Least-squares fit on 495 live points: $w_{\text{cost}} = 0.500025$, $w_{\text{nlp}} = 0.499989$ (Max error ≤ 0.05).

SINGLE CITIZEN BOOST

+15.0 is awarded on the FIRST report only. Subsequent reports increment count without inflating score.

FALSE POSITIVE GUARD

When an officer records False Positive, score drops by 25 points and dampens NLP twins by 10 points.

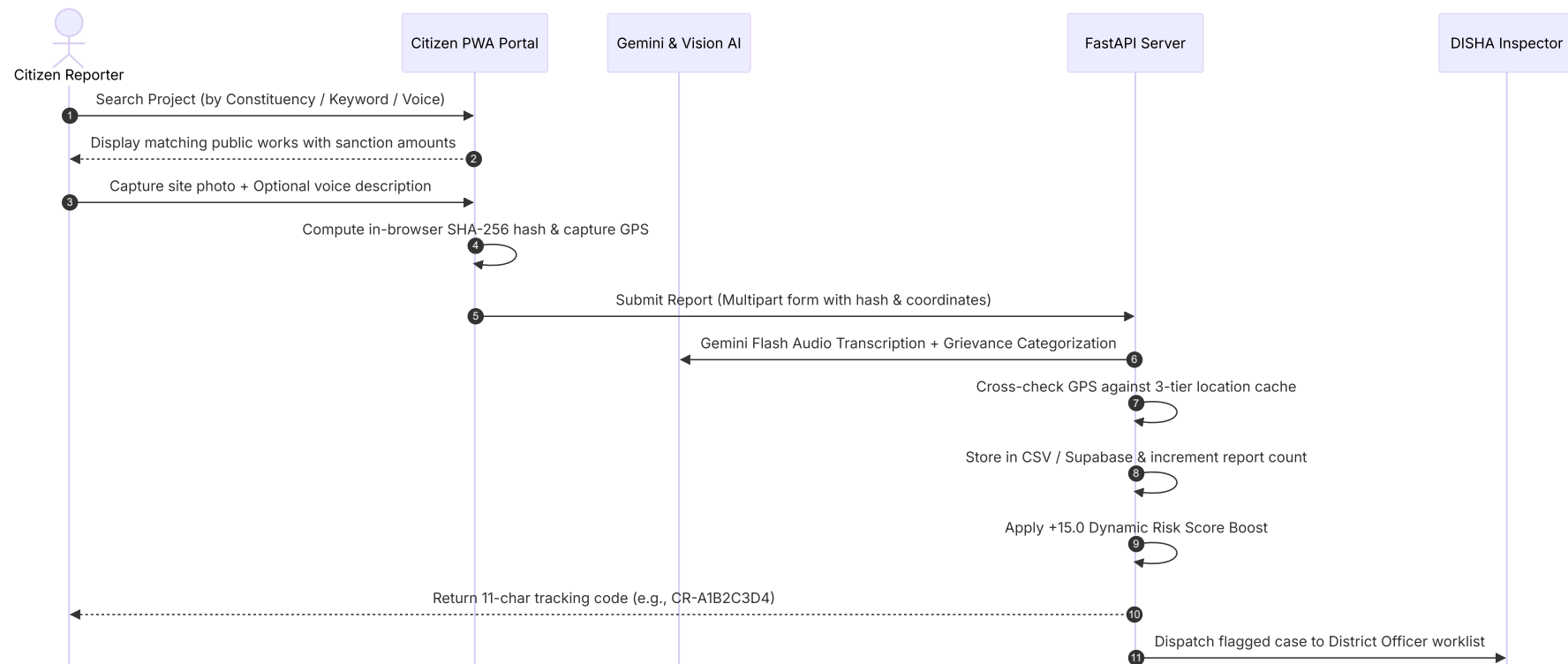
EXPLAINABILITY LAYER

Gemini Flash provides plain-language justification using read-only inputs without modifying scores.

3. Citizen Grievance Intake & AI Cross-Verification Sequence

Client-side cryptographic fingerprinting, audio transcription, and dynamic score dispatching

Sequence Workflow



PRECISE GPS TIER
100m – 200m radial tolerance for exact worksite coordinates. Full 30% verification weight.

LOCALITY REGEX TIER
2km primary tolerance (10km ceiling) for village/panchayat extracted from project description.

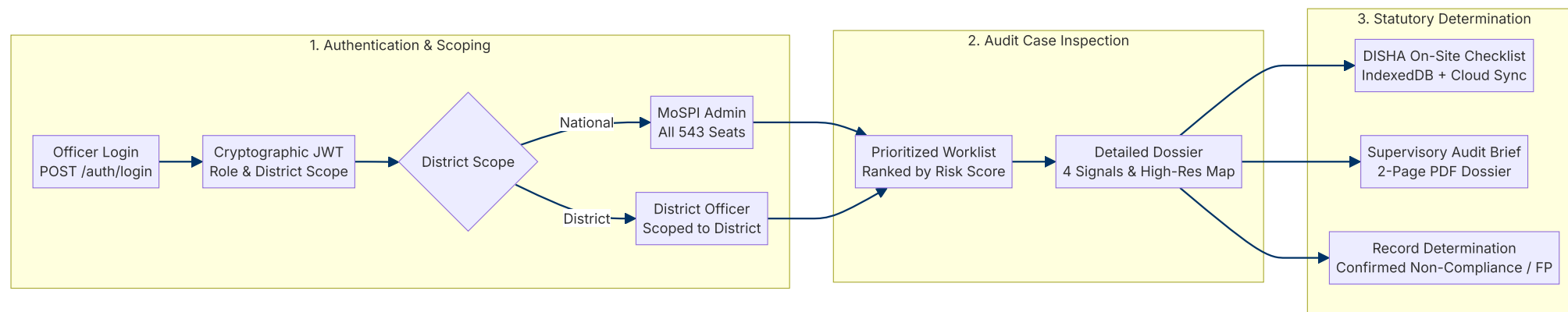
DISTRICT CENTROID TIER
25km primary tolerance. Adapts weight (30% → 20%), shifting 10% to photo and text verification.

EVIDENCE FINGERPRINTING
Browser Web Crypto SHA-256 hash prevents tampering; verified against uploaded payload.

4. DISHA Officer Workflow & Jurisdiction Scoping

GIGW 3.0 compliant workspace for district inspection officers and central auditors

Officer Lifecycle



JURISDICTION SCOPING

Officers cannot view records outside statutory district boundary; strictly enforced server-side.

DISHA DIGITAL CHECKLIST

10-point statutory physical verification form with offline caching and Supabase synchronization.

2-PAGE AUDIT BRIEF

ReportLab-generated PDF dossier with Sentinel-2 aerial thumbnail, cost Z-score, and Gemini notes.

STATUTORY COMPLIANCE

Adheres to Information Technology Act 2000 • DPDP Act 2023 • GIGW 3.0 Government Standards.